COURSE DESCRIPTION

1. Program Information

1.1 University	**“Alexandru Ioan Cuza” University of Iaşi**
1.2 Faculty	**Faculty of Computer Science**
1.3 Department	**Department of Computer Science**
1.4 Study Domain	**Computer Science**
1.5 Study Cycle	**Undergraduate studies**
1.6 Study Program / Qualification	**Computer Science/Computer Science Graduate**

2. Course Information

2.1 Course Name			**Quantum Computing**				
2.2 Course Teacher			Conf. dr. Andreea-Valentina Arusoaie				
2.3 Seminary Teacher			Conf. dr. Andreea-Valentina Arusoaie				
2.4 Year of study	**3**	**2.5** Semester	**1**	**2.6** Evaluation	**E**	2.7 Course status*	**OP**

* OB – Obligatoriu / OP – Opţional

3. Total estimated hours (hours per semester and didactic activities)

3.1 Hours per week	**4**	in which: **3.2** course	**2**	**3.3** seminary/laboratory	**2**
3.4 Hours in curriculum	**56**	in which: **3.5** course	**28**	**3.6** seminary/laboratory	**28**

Time Distribution	hours
Manual study, Course support, Bibliography, and others	**10**
Supplementary Documentation in library, in electronic forums, and on the field	**15**
Seminaries/laboratories preparation, homeworks, reports, portfolios and essays	**15**
Tutoring	**0**
Evaluation	**4**
Other activities (consultations per student)	**0**

3.7 Total hours individual study	**40**
3.8 Total hours per semester	**100**
3.9 Credits	**4**

4. Preconditions (if necessary)

4.1 Curriculum	**Computer Architecture and Operating Systems, Mathematics, Algorithm Design**
4.2 Competencies	**Design logical gates, Imperative programming**

5. Conditions (if necessary)

5.1 For Course Operation	- Class with video-projector / laptop/PC with good internet connection
5.2 For Seminary/Laboratory Operation	- Class with video-projector / laptop/PC with good internet connection

6. Specific Skills Acquired

Professional Skills	**C1.** The understanding and handling of specific concepts related to Linear Algebra and Mathematical Analysis. **C2.** The understanding and handling of basic notions related to Quantum Physics. **C3.** The application of learned mathematical notions for solving practical problems of quantum computing. **C4.** Use a quantum framework to implement a quantum circuit, quantum algorithm.
Transversal Skills	**C1.** Capacity to recognize some problems that can be solved using a quantum computer. **C2.** Capacity to design some quantum circuits and implement some quantum algorithms.

7. Course Objectives (from the grid of specific skills acquired)

7.1 General Objective	Presenting some basic notions of quantum computing.
7.2 Specific Objectives	After taking this course, students will be able to : O1. Recognize some problems that can be solved using quantum computing O2. Explain concepts specific to quantum computing O3. Design quantum circuits for solving practical problems O4. Design a quantum algorithm

8. General Description

8.1	**Lecture**	**Teaching method**	**Remarks** (hours, references)
1.	Introduction to Quantum Computing.	Exposition, Debates, Demonstration	2 hours, [1],[2]
2.	Mathematical foundations: Notions on linear Algebra and Mathematical Analysis specific for Quantum Computing	Exposition, Debates, Demonstration	2 hours, [1], [2], [3]
3.	The Postulates of Quantum Mechanics	Exposition, Debates, Demonstration	2 hours, [1], [2], [3]

4.	Quantum computing: Logical Gates. 1-Qubit Gates. Examples	Exposition, Debates, Demonstration	2 hours, [1], [2], [4]
5.	Quantum circuits: Multiple qubit gates	Exposition, Debates, Demonstration	2 hours, [1], [2],
6.	Quantum Information: No Cloning theorem. Entanglement.	Exposition, Debates, Demonstration	2 hours [1], [2],
7.	Quantum key distribution. Quantum Teleportation	Exposition, Debates, Demonstration	2 hours [1], [2],
8.	Review of the notions exposed in the first seven lectures	Written test	2 hours
9.	Quantum algorithms: Deusch-Jozsa Algorithm	Exposition, Debates, Demonstration	2 hours [1], [3]
10.	Quantum algorithms: Bernstein-Vazirani. Simon’s Algorithm	Exposition, Debates, Demonstration	2 hours, [1], [2]
11.	Quantum Search	Exposition, Debates, Demonstration	2 hours [1], [2], [3]
12.	Fourier transform: discrete version, quantum version. Phase estimation	Exposition, Debates, Demonstration	2 hours, [1], [2], [3]
13.	Quantum Fourier Transform: Factorization Problem	Exposition, Debates, Demonstration	2 hours, [1], [2], [3]
14.	Quantum computing: Open problems, conclusions.	Exposition, Debates, Demonstration	2 hours

References

[1] M.Nielsen and I.Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2000.

[2] P. Kaye, R. Laflamme, and M. Mosca, *An Introduction to Quantum Computing*, Oxford University Press, 2007.

[3] A. Kitaev, A. Shen, and M. Vyalyi, *Classical and Quantum Computation*, Volume 47 of Graduate Studies in Mathematics. American Mathematical Society, 2002.

[4] Chris Bernhardt, Quantum computing for everyone, Massachusetts Institute of Technology, 2019

8.2	**Seminar / Laboratory**	**Teaching methods**	**Observations** (hours and bibliography)
1.	Introduction to Quantum Computing. Bit. Qubit	Exercise, conversations	2 hours [1],[2]
2.	Exercise sheet: Mathematical foundations	Exercise, conversations	2 hours [1],[2]

3.	Exercise sheet: Quantum bits, Superposition, Entanglement	Exercise, conversations	2 hours [1],[2]
4.	Exercise sheet: Quantum bits, Superposition, Entanglement	Exercise, conversations	2 hours [2]
5.	Exercise sheet: Quantum circuits: Logical Gates, Examples	Exercise, conversations	2 hours [1]
6.	Exercise sheet: Quantum computations	Exercise, conversations	2 hours [2]
7.	Exercise sheet: Introduction in Python	Exercise, conversations	2 hours [3]
8.	Review of the notions exposed in the first seven lectures	Evaluation	2 hours
9.	Exercise sheet: implementing quantum algorithms	Exercise, conversations	2 hours [2],[3]
10.	Exercise sheet: implementing quantum algorithms	Exercise, conversations	2 hours [3].[4]
11.	Exercise sheet: implementing quantum algorithms.	Student presentation and discussion.	2 hours [2],[3]
12.	Exercise sheet: implementing quantum algorithms. Evaluation of assignements	Student presentation and discussion.	2 hours [3]
13.	Exercise sheet: implementing quantum algorithms. Evaluation of assignements.	Student presentation and discussion.	2 hours [3]
14.	Exercise sheet: implementing quantum algorithms. Evaluation of assignements	Student presentation and discussion.	2 hours [4]

References

[1] N.S. Yanofsky, M.A. Mannucci, *Quantum Computing for Computer Scientists,* Cambridge University Press, 2008.
[2] A. Kitaev, A. Shen, and M. Vyalyi, *Classical and Quantum Computation*, Volume 47 of Graduate Studies in Mathematics. American Mathematical Society, 2002.
[3] H. Norlen - Quantum Computing in Practice with Qiskit and IBM Quantum Experience, Packt Publishing, 2020.
[4] IBM quantum experience: https://quantum-computing.ibm.com/

9. Course content synchronization with the expectations of the community representatives, professional associations and employers from the program domain

This discipline aims to present some basic notions in the domain of Quantum Computing. The students will learn to identify and solve practical problems using quantum computing. Also, the students will learn to use the IBM quantum computer.

10. Evaluation

Activity Type	**10.1.1 Evaluation criteria**	**10.2 Evaluation methods**	**10.3 The weight of each evaluation form (%)**
10.4 Course	60 points	Exam - week 8	60%
10.5 Seminary/ Laboratory	40 points	Lab/homework assignments	40%
10.6 Minimum requirements for graduation : ● Design a quantum circuit in order to solve a practical problem. ● to use a quantum circuit simulator and to implement quantum algorithms. ● define the notion of qubit, superposition, and entanglement.			
In order to meet the minimum requirements, it is recommended to participate at the evaluation - week 8 and present the lab assignments. Minimum requirements: Evaluation grade>=4.5. Total score of at least 4.5. The presentation of the assignments will be done in the second part of the semester.			

Date
20/09/2024

Course and Seminary/Laboratory Teacher
Conf. dr. Andreea-Valentina Arusoaie

Department Date of Approval

Director of the Department
Conf. dr. Andrei Arusoaie